

Why Ethics?

2021-07-21

Potential Jobs for Computer and Data Scientists

Here is a list of potential jobs in the computer and data sciences.

Computer Science

- Software developer
 - Games
 - Social media
 - Airplane/Car navigation systems
 - Predictive algorithms (who is a good candidate for a loan or job, predicting future weather or erosion patterns, predicting if a person who is arrested is a flight risk)
 - Science applications (finding genetically-related diseases, analyzing tumor progression/reduction, drug discovery)
 - Compilers/operating systems
- Database Administrator
 - Design systems to store and retrieve important information
- Computer hardware engineer
 - Design the next phone, computer or tablet
 - Create/test circuit boards, routers, memory devices
 - Create ways for disabled to move and/or communicate
- Information security analyst
 - Detect cyber attacks
 - Prevent cyber attacks
- Web Developer
 - Create the technical structure for websites
 - Make sure that web pages are accessible and easily downloadable

- Structure sites to maximize the number of page views and visitors through search engine optimization.

Data Science and Statistics

- Data Analytics
 - Help develop strategies to make better business decisions
 - Create/use programs to help individuals invest money
 - Track and analyze website usage patterns
- Data architect
 - Find, clean and organize data
 - Look for patterns in data collected
- Machine learning
 - Develop predictive algorithms based on what has happened in the past (medicine, business, job/student success, economy, climate change, disease spread, loan repayment, criminal detainee flight risk etc.)
- Statistician
 - Interpret, analyze, and report statistical information (could be related to research in any field: business, psychology, sociology, medicine..)

Group work

In the next 10 minutes do the following in your group:

- Choose a note-taker and spokesperson so that your ideas can be shared with the larger group.
- Brainstorm with your group things that could go wrong in any of these areas if they are not handled carefully.
- Brainstorm with your group the things that could be beneficial to society if they were handled in the best possible way.

Example

A computer hardware designer could create a new phone that was feature rich but also designed to be obsolete within a year so that the user would need to replace it quickly. This would allow for a significant revenue stream for the company. However, the plethora of features could be too complex for an elderly person to manage, and the planned obsolescence could result in excessive waste.

On the flip-side, the designer could implement a version of the phone where the initial setup included a choice of “essentials only” resulting in a screen that was easy to navigate and included a large, clean, display. There could be a prominent “help” button that connected the user with a predetermined relative or caregiver. The phone could come with an offer of a discount if the buyer’s previous phone was turned in for recycling (materials reuse) or donation to a non-profit.

Bibliography